

CHAPTER 2

MATRIX ALGEBRA

2.2. MATRIX ADDITION, SCALAR MULTIPLICATION, AND TRANSPOSITION

In this section we discuss the main components of matrices and describe three fundamental operations on them.

A **matrix** is a two-dimensional array of numbers. We usually denote matrices with uppercase letters.

The "anatomy" of a matrix is as follows

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1(n-1)} & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2(n-1)} & a_{2n} \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ a_{(m-1)1} & a_{(m-1)2} & \cdots & a_{(m-1)(n-1)} & a_{(m-1)n} \\ a_{m1} & a_{m2} & \cdots & a_{m(n-1)} & a_{mn} \end{bmatrix}$$

Annotations:

- Entry (there are $m \cdot n$ of them)
- Row (there are m of them)
- Column (there are n of them)

For convenience, we simply write $A = [a_{ij}]$

SOME TERMINOLOGY

- The **(i,j) -entry** of A is the entry appearing in the i -th row and j -th column.
- If A has m rows and n columns, we say that it has **size $m \times n$** (or that it is an **$m \times n$ matrix**).
- An $1 \times n$ matrix is called a **row matrix**; an $l \times n$ matrix is called a **column matrix**.
- An $n \times n$ matrix is called a **square matrix**.

We will say that two matrices $A = [a_{ij}]$ and $B = [b_{ij}]$ are **equal** ($A=B$) if they have the same size and $a_{ij} = b_{ij}$ for all i and j .

MATRIX ADDITION

Suppose $A = [a_{ij}]$ and $B = [b_{ij}]$ are two matrices of the same size. We define their sum $A+B$ as the matrix

$$A+B = [a_{ij} + b_{ij}].$$

For example:

$$\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix} + \begin{bmatrix} 0 & -1 & 7 \\ 8 & -10 & -2 \end{bmatrix} =$$

REMEMBER: $A+B$ only makes sense if A and B have the same size

The $m \times n$ zero matrix is the matrix O with all entries equal to zero. This matrix satisfies $O+X = X+O = X$ for all $m \times n$ matrices X . Furthermore, the negative of a matrix $A = [a_{ij}]$ is the matrix $-A = [-a_{ij}]$. This matrix satisfies $A+(-A) = (-A)+A = O$. The difference of two $m \times n$ matrices A and B is $A-B = A+(-B)$.

For example:

$$\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix} - \begin{bmatrix} 0 & -1 & 7 \\ 8 & -10 & -2 \end{bmatrix} =$$

SCALAR MULTIPLICATION

Suppose $A = [a_{ij}]$ is a matrix and k is any number (we also say that k is a scalar). The scalar multiple kA is the matrix

$$kA = [ka_{ij}]$$

For instance, $1 \cdot A = A$ and $0 \cdot A = O$.

THEOREM: Let A, B and C be $m \times n$ matrices, and k, p two scalars. Then the following identities hold:

1) $A+B = B+A$ (Commutative law)

5) $k(A+B) = kA + kB$ (Distributive law 1)

2) $A+(B+C) = (A+B)+C$ (Associative law)

6) $(k+p)A = kA + pA$ (Distributive law 2)

3) $O+A = A+O = A$

7) $(kp)A = k(pA)$

4) $A+(-A) = (-A)+A = O$

8) $1A = A$

PROOF

We write $A = [a_{ij}]$ and $B = [b_{ij}]$. We prove the identities one by one.

$$1) A+B = [a_{ij} + b_{ij}] = [b_{ij} + a_{ij}] = B+A$$

$$2) A+(B+C) =$$

$$3) 0+A = [0 + a_{ij}] = [a_{ij}] = A = [a_{ij} + 0] = A+0.$$

$$4) A+(-A) = [a_{ij}] + [-a_{ij}] = [a_{ij} - a_{ij}] = [0] = 0. \text{ Also, } (-A)+A = A+(-A) = 0 \text{ (Using the commutative law).}$$

$$5) k(A+B) =$$

$$6) (k+p)A = [(k+p)a_{ij}] = [ka_{ij} + pa_{ij}] = [ka_{ij}] + [pa_{ij}] = kA + pA.$$

$$7) (kp)A =$$

$$8) 1A = //$$

EXAMPLE: Simplify the expression $2[7(A-C)] + \frac{1}{2}B + 9[A-B+3C]$

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EXAMPLE: Find the matrix X that satisfies

$$2X + \begin{bmatrix} 1 & 7 & -8 \\ 6 & 5 & 4 \end{bmatrix} = \begin{bmatrix} 0 & 2 & 0 \\ 4 & 0 & 0 \end{bmatrix} - X$$

SOL:

TRANSPOSE OF A MATRIX

Suppose A is an $m \times n$ matrix. We define the transpose A^T of A as the $n \times m$ matrix.

$$A^T = [a_{ji}]$$

That is, the rows of A^T are the columns of A in the same order. For instance:

$$\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}^T = \begin{bmatrix} 1 & 4 \\ 2 & 5 \\ 3 & 6 \end{bmatrix}.$$

One can think of A^T as the result of "reflecting" A along its main diagonal — that is, the diagonal consisting of $a_{11}, a_{22}, a_{33}, \dots$

$$\begin{bmatrix} 4 & 2 & 1 & 0 \\ 1 & 1 & -7 & 6 \end{bmatrix}^T = \begin{bmatrix} 4 & 1 \\ 2 & 1 \\ 1 & -7 \\ 0 & 6 \end{bmatrix}$$

Main diagonal

A square matrix is symmetric if $A^T = A$.

THEOREM: If A and B are matrices of the same size and K is a scalar, then:

1) $(A^T)^T = A$.

2) $(kA)^T = kA^T$.

3) $(A+B)^T = A^T + B^T$

(EXERCISE: Verify these formulas.)

EXAMPLE: Find the matrix X that satisfies

$$\left(2A^T - 3 \begin{bmatrix} 1 & 2 \\ -1 & 1 \end{bmatrix} \right)^T = \begin{bmatrix} 2 & 3 \\ -1 & 2 \end{bmatrix}$$

SOL:

EXERCISE: If A is a matrix and k is a number such that $kA = 0$, show that $k=0$ or $A=0$.

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